

The Epistemic Price of Coordination: Minimal Handshakes under Private Game Models

Anonymous submission

Abstract

Cooperative agents deployed by different developers may face the same task without sharing a common model of its payoffs, action semantics, or observation process. We formulate *epistemic coordination games*, finite common-payoff games in which each agent observes only a private game-model type and one agent may send a bounded pre-play handshake. We define the epistemic price of coordination (EPoC) as the minimum worst-case regret relative to a world-aware oracle, separating communication-repairable convention mismatch from uncertainty that no message can remove. At zero communication, threshold feasibility is a CSP: it reduces to linear-time 2-SAT for binary actions, yet is NP-complete with three actions even for binary rewards. For one-way communication, the exact message requirement is the chromatic number of a compatibility hypergraph. Pairwise confusability is insufficient: a projective-plane family has an empty pairwise conflict graph but requires $q + 1$ messages, an unbounded gap. We also identify positive structure through Helly bounds, exact interval stabbing, and bounded-treewidth dynamic programming. Metric action-dictionary games reduce exactly to robust M -center. On odd cycles, communication quantization, sender target uncertainty, and latent receiver-dictionary uncertainty add in a closed form. Exact experiments, including exhaustive small-cycle checks and rotated/reflected 2D maps, verify the predictions and expose irreducible uncertainty floors.

Introduction

Zero-shot cooperation asks independently designed agents to coordinate without joint training or prior interaction. Existing methods often seek policies that are invariant to arbitrary conventions or robust across partner populations [Hu et al. 2020, Strouse et al. 2021, Li et al. 2023, Jha et al. 2025]. Yet a distinct failure occurs before strategic reasoning begins: agents may not agree on the game they are playing. One robot’s “north” may be another’s “east”; one planner may believe a role is feasible while its partner does not; two agents may receive different noisy payoff tables; or they may disagree about which facts are public. Recent noisy zero-shot coordination explicitly models agents with different private

environment descriptions, but retains common knowledge of the process producing those descriptions [Anwar et al. 2024]. In heterogeneous deployments, even that shared model may be unavailable.

We study the smallest setting in which this epistemic mismatch can be isolated from learning: a finite two-agent common-payoff game, private local descriptions of the realized game, and a bounded one-way pre-play message. The central question is not whether unrestricted communication can reveal a type. It is: *which distinctions between private models matter strategically, how many handshake symbols repair them, and what performance remains impossible because neither agent knows the missing fact?*

Our first object is a loss–communication frontier. For a message alphabet of size M , $\text{EPoC}(M)$ is the least worst-case regret of any deterministic one-way protocol relative to an oracle that observes the realized world. Its inverse $C(\epsilon)$ is the minimum number of messages needed to guarantee regret at most ϵ . Unlike communication cost alone, the frontier retains the game payoffs: two model distinctions may be informationally different but strategically interchangeable at a given tolerance.

Our contributions are fourfold.

1. We give a finite-game formulation of EPoC and show a sharp action-cardinality boundary for zero communication: threshold feasibility reduces to 2-SAT with two actions, but is NP-complete with three.
2. We characterize bounded one-way handshakes by a *compatibility hypergraph*. This higher-order object is necessary: using projective planes, we construct instances in which every pair of private models can share one message, while exact coordination needs $q + 1$ messages.
3. We identify tractable structure. Convex response sets imply hyperedges of size at most $d + 1$; intervals reduce exactly to pairwise conflicts and admit optimal right-endpoint greedy stabbing; and fixed-message feasibility is polynomial on bounded-treewidth type graphs for fixed message and action alphabets.
4. We reduce metric action-dictionary games exactly to robust M -center and derive an additive odd-cycle frontier: center quantization, sender-side target uncertainty,

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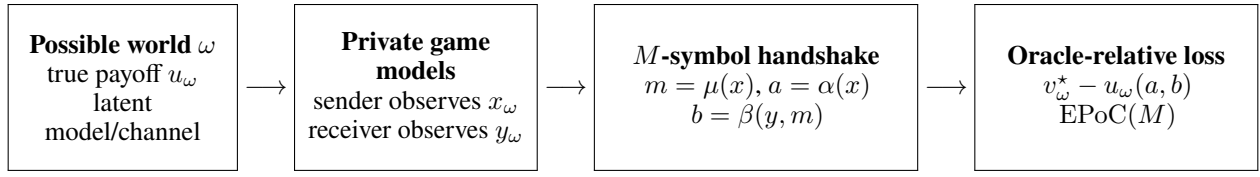


Figure 1: **Epistemic coordination.** Agents cooperate in the same true world but receive only private descriptions of it. A bounded pre-play message can repair distinctions between local models only when those distinctions can be compressed into strategically compatible classes.

and latent receiver-dictionary uncertainty contribute separately before the diameter cap. A 2D map benchmark and a general full-revelation theorem separate repairable mismatch from absent information.

Our formulation intersects classical zero-error coordination and side-information coding [McEliece and Posner 1971, Witsenhausen 1976, Abroshan et al. 2015]. We do not claim generic hitting-set or graph-coloring reductions as new. Instead, the contribution is a game-regret frontier for private game models, together with sharp structural results showing when pairwise abstractions fail, when they become exact, and how communication separates convention mismatch from irreducible uncertainty.

Epistemic Coordination Games

A finite epistemic coordination instance is

$$\mathcal{I} = (\Omega, X, Y, A, B, \tau_1, \tau_2, (u_\omega)_{\omega \in \Omega}).$$

A world $\omega \in \Omega$ specifies a common payoff $u_\omega : A \times B \rightarrow \mathbb{R}$. The sender observes only $x_\omega = \tau_1(\omega) \in X$ and the receiver observes only $y_\omega = \tau_2(\omega) \in Y$. A type can encode a local payoff table, map, action dictionary, capability model, or observation model; different worlds may induce the same local type.

An M -message deterministic one-way protocol is

$$\pi = (\mu, \alpha, \beta), \quad \mu : X \rightarrow [M], \quad \alpha : X \rightarrow A, \\ \beta : Y \times [M] \rightarrow B.$$

Thus the sender transmits $\mu(x)$ and acts with $\alpha(x)$; the receiver acts with $\beta(y, \mu(x))$. Let $v_\omega^* = \max_{a,b} u_\omega(a, b)$. The protocol's worst-case regret is

$$R(\pi) = \max_{\omega \in \Omega} [v_\omega^* - u_\omega(\alpha(x_\omega), \beta(y_\omega, \mu(x_\omega)))].$$

Definition 1 (Epistemic price and communication threshold).

$$\text{EPoC}_{\mathcal{I}}(M) = \min_{\pi: |\text{range}(\mu)| \leq M} R(\pi), \\ C_{\mathcal{I}}(\epsilon) = \min\{M : \text{EPoC}_{\mathcal{I}}(M) \leq \epsilon\}.$$

We use the convention $\min \emptyset = \infty$. A budget of b bits permits at most 2^b messages.

Because the instance is finite, the minimum is attained and $\text{EPoC}(M)$ is a nonincreasing staircase whose values come from the finite set of world/action regrets. We use deterministic robust protocols.

Why deterministic protocols? A public-randomized protocol is a distribution p over deterministic M -message protocols, evaluated by worst-case expected regret

$$\text{EPoC}^{\text{pub}}(M) = \min_p \max_{\omega} \mathbb{E}_{\pi \sim p} [r_\omega(\pi)].$$

Randomization can improve a positive-regret frontier. For example, let two indistinguishable worlds prefer opposite receiver actions with binary payoff. Every deterministic choice has worst-case regret one, whereas a fair public coin has expected regret $1/2$ in both worlds. It cannot, however, create exact coordination.

Proposition 2 (Zero-regret derandomization). *A public-randomized M -message protocol has zero expected regret in every world if and only if a deterministic M -message zero-regret protocol exists.*

Proof. Nonnegative regret with expectation zero is zero almost surely in each world. The finite intersection of these probability-one events contains a common seed, which induces a deterministic protocol with zero regret everywhere. The reverse direction is immediate. $\square$

We focus on deterministic protocols because they capture standardized interoperability handshakes, preserve an exact combinatorial message meaning, and admit sharp structural characterizations. The randomized positive-regret frontier is an important extension rather than a hidden assumption.

For tolerance ϵ , define the acceptable relation

$$\mathcal{R}_\omega^\epsilon = \{(a, b) : v_\omega^* - u_\omega(a, b) \leq \epsilon\}.$$

All feasibility results below depend only on these relations, while the full EPoC frontier is obtained by searching the finitely many regret thresholds. Table 1 previews how the correct combinatorial object changes with the information and game structure.

Zero Communication: A Sharp Boundary

With one message, the agents choose local rules a_x and b_y . Hence threshold feasibility is exactly the binary CSP with variables $A_x \in A$, $B_y \in B$, and constraints $(A_{x_\omega}, B_{y_\omega}) \in \mathcal{R}_\omega^\epsilon$. This elementary equivalence becomes useful through the following boundary.

Theorem 3 (Action-cardinality boundary). *Zero-communication ϵ -coordination is decidable in linear time after scanning the payoff tables when both agents have at most two actions. With three actions per agent, it is NP-complete even for $\epsilon = 0$ and payoffs in $\{0, 1\}$.*

Setting	Exact object	Consequence
$M = 1$, binary actions	2-SAT	linear time
$M = 1$, three actions	finite CSP	NP-complete
General M	compatibility hypergraph	$C(\epsilon) = \chi(H_\epsilon)$
One receiver, binary payoff	hitting set	NP-hard; greedy bound
Convex responses in $\mathbb{R}^d$	rank- $d+1$ hypergraph	small obstruction certificates
Interval responses	interval stabbing	greedy exact
Metric action dictionaries	robust M -center	exact reduction; 2-approx.
Cyclic hidden dictionary bias	cycle covering	exact additive frontier
Type graph of width w	binary CSP	polynomial for fixed $M, w, A , B $

Table 1: Structural landscape of deterministic one-way episodic coordination.

Proof idea. For binary actions, every forbidden pair (a, b) in a world yields the clause $(A_x \neq a) \vee (B_y \neq b)$, giving at most four 2-SAT clauses per world. For hardness, reduce graph 3-coloring. Create sender and receiver copies x_v, y_v of every vertex. Equality worlds force their actions to match; one inequality world for each oriented edge forces adjacent colors to differ. Full proofs are in the supplement.

The theorem shows that computational difficulty can arise solely from inconsistent local game models: the underlying game is one-shot, cooperative, and uses deterministic binary rewards.

Minimal Handshakes Need Hypergraphs

Fix ϵ . A set $S \subseteq X$ is *compatible* if all its sender types can share one message: there exist sender actions $(a_x)_{x \in S}$ and receiver responses $(b_y)_{y \in Y}$ such that $(a_x, b_{y_\omega}) \in \mathcal{R}_\omega^\epsilon$ for every world whose sender type lies in S .

Let $H_\epsilon = (X, \mathcal{E}_\epsilon)$ contain as hyperedges the inclusion-minimal incompatible subsets of sender types. A proper hypergraph coloring assigns no hyperedge a single color; if a singleton hyperedge exists, we set $\chi(H_\epsilon) = \infty$.

Theorem 4 (Compatibility characterization). *The exact message requirement is*

$$C_T(\epsilon) = \chi(H_\epsilon).$$

Proof. Every message fiber of a feasible protocol is compatible, so its message map is a proper coloring. Conversely, every color class of a proper coloring must be compatible: otherwise it contains a minimal incompatible hyperedge. Choosing a compatibility witness for each color yields a protocol. $\square$

An ordinary pairwise conflict graph records only incompatible pairs. It can miss obstructions involving three or more private models. The smallest example has one receiver type and acceptable receiver-action sets $\{A, B\}, \{B, C\}, \{A, C\}$. Each pair intersects, but the triple does not; one message fails while two suffice.

The gap can grow without bound.

Theorem 5 (Unbounded pairwise gap). *For every prime-power projective-plane order q , there is a binary-payoff instance with $q^2 + q + 1$ sender types such that every pair is compatible, but exact coordination requires exactly $q + 1$ messages.*

Construction and proof. Use the lines of $\text{PG}(2, q)$ as sender types, its points as receiver actions, one receiver type, and payoff one when the selected point lies on the sender's line. Two lines always intersect, so the pairwise graph is empty. One message class must consist of lines sharing a selected point; therefore the minimum messages equal the minimum number of points hitting all lines. The $q + 1$ points on one line hit every line. Conversely, for any set P of at most q points, choose $p \notin P$. Of the $q + 1$ lines through p , each point in P hits only one, leaving a line unhit. Thus $q + 1$ messages are necessary. $\square$

Theorem 5 gives an unbounded multiplicative separation between pairwise analysis (one message) and the true requirement. It also explains why a protocol designer should cluster model types by *joint* strategic compatibility, not by pairwise similarity.

Algorithms and Positive Structure

Exact finite solver. For fixed M and ϵ , introduce binary variables s_{xma} indicating that sender type x emits message m and takes action a , and r_{ymb} indicating receiver response b . We solve

$$\sum_{m,a} s_{xma} = 1 \quad \forall x, \quad (1)$$

$$\sum_b r_{ymb} = 1 \quad \forall y, m, \quad (2)$$

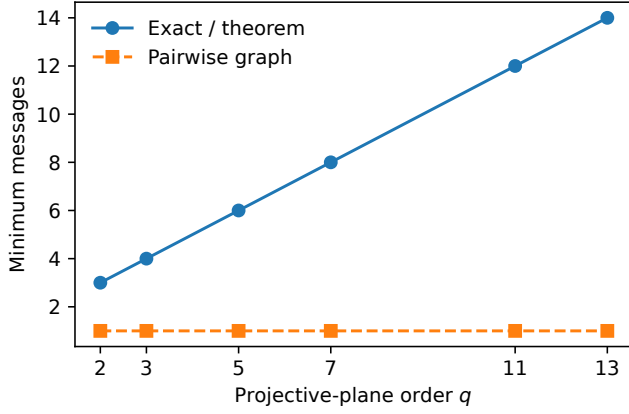
$$s_{x_\omega ma} + r_{y_\omega mb} \leq 1 \quad \forall \omega, m, (a, b) \notin \mathcal{R}_\omega^\epsilon. \quad (3)$$

The formulation has $|X|M|A| + |Y|M|B|$ variables and at most $|X| + |Y|M + |\Omega|M|A||B|$ constraints.

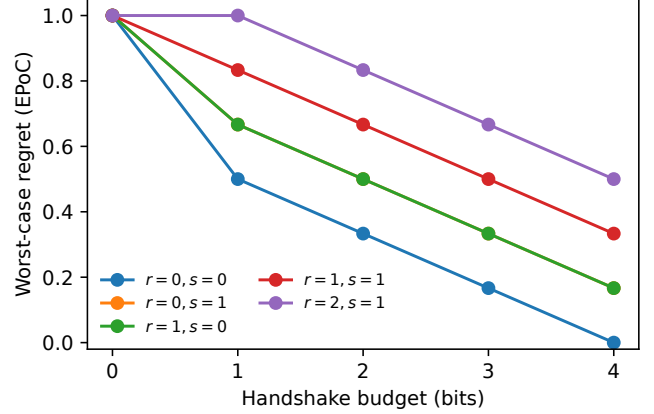
Proposition 6 (MILP correctness). *Equations (1)–(3) are feasible if and only if an M -message deterministic protocol has regret at most ϵ . Searching the finite regret levels returns exact $\text{EPoC}(M)$.*

The proof reads an integral solution as local rules and conversely encodes any protocol. We independently decode and reevaluate every returned protocol. The MILP is intended for validation and small design problems, not as a claim of general polynomial-time tractability.

One receiver and set systems. When the sender action is trivial, there is one receiver type, rewards are binary, and sender type x accepts a set $S_x \subseteq B$, exact coordination reduces to selecting receiver actions that hit every S_x . Thus $C(0)$ is the transversal number. This specialization connects our framework to classical one-shot zero-error coordination and set cover [McEliece and Posner 1971, Abroshan et al. 2015, Chvatal 1979, Feige 1998]; we use it for transparent exact and greedy baselines rather than claim the reduction as new. It immediately implies that exact message minimization is NP-hard even in this restricted cooperative setting. The standard greedy rule that repeatedly chooses the receiver



(a) Higher-order compatibility gap



(b) Loss-communication frontiers

Figure 2: **Exact theoretical predictions.** (a) In projective-plane games, the pairwise conflict graph predicts one message while exact MILP and Theorem 5 require $q + 1$. (b) In the 13-lever game, communication quantization, sender uncertainty r , and hidden dictionary uncertainty s add until the physical diameter cap.

action compatible with the most uncovered sender types returns at most $H_{|X|}$ times the optimal number of messages [Chvatal 1979]; classical set-cover lower bounds also transfer [Feige 1998]. These are inherited approximation statements, but they provide a principled baseline for the game-design problem.

Corollary 7 (Message minimization in set systems). *Minimum-message exact coordination is NP-hard even with one receiver type, a trivial sender action, and binary rewards. Greedy returns at most $H_{|X|}C(0)$ messages in this subclass.*

The projective construction shows unrestricted set systems require genuinely higher-order reasoning. Geometric structure restores locality.

Theorem 8 (Helly rank). *Consider a receiver-only instance with responses in $\mathbb{R}^d$. If every acceptable response set for a sender/receiver type pair is convex, every minimal incompatible sender set has size at most $d + 1$.*

Proof. If a sender class is incompatible, then for some receiver type the corresponding convex acceptable sets have empty intersection. Helly’s theorem gives a subfamily of at most $d + 1$ with empty intersection. A minimal incompatible set can be no larger. $\square$

For intervals ($d = 1$), pairwise compatibility is therefore sufficient. With one receiver type, the minimum messages equal the minimum number of points stabbing all acceptable intervals, found exactly by repeatedly selecting the right endpoint of the earliest-ending unhit interval. This contrast precisely locates a regime in which pairwise graphs are sound.

Sparse epistemic interaction. Merge worlds with the same type pair into one acceptable relation and form the bipartite type-interaction graph on $X \cup Y$. For fixed M , assign each sender variable a pair $(m, a) \in [M] \times A$ and each receiver variable a response vector in B^M . This is a binary CSP on the type graph.

Proposition 9 (Bounded-treewidth algorithm). *Given a width- w tree decomposition, fixed- M threshold feasibility is solvable*

in

$$O((|X| + |Y|)D^{w+1}), \quad D = \max\{M|A|, |B|^M\},$$

up to polynomial local-constraint checks.

This dependence is practical for the intended regime of one- to three-bit handshakes when private-model interactions are sparse.

Metric Action Dictionaries

Many interoperability problems have a shared physical goal space but private coordinate systems or action dictionaries. Let (V, d) be a finite metric space with diameter D . The sender observes an estimate $x \in X \subseteq V$; the true target belongs to a nonempty uncertainty set $U_x \subseteq V$. Receiver type g is a bijective isometry from local actions V to physical outcomes V . If the receiver chooses local action b , the payoff is $1 - d(g(b), t)/D$ for true target $t \in U_x$.

Define the robust assignment cost

$$\Delta(c, x) = \max_{t \in U_x} d(c, t).$$

Theorem 10 (Metric dictionary reduction). *For every M ,*

$$\text{EPoC}(M) = \frac{1}{D} \min_{c_1, \dots, c_M \in V} \max_{\substack{x \in X \\ \mu: X \rightarrow [M]}} \Delta(c_{\mu(x)}, x).$$

Thus epistemic protocol design in this family is exactly a robust metric M -center problem. When $U_x = \{x\}$, it is ordinary metric M -center and farthest-first traversal gives a 2-approximation [Gonzalez 1985].

Proof. For message m and receiver isometry g , any local response corresponds to a physical center $c_{g,m} = g(b)$. Because g is bijective, every physical center is available. The target sets do not depend on g , so each receiver type faces the same minimax center problem for a message class and may implement a common optimal physical center through g^{-1} . Optimizing the message partition and these centers gives the display. $\square$

The reduction distinguishes two issues. A private coordinate frame causes no additional loss once a message identifies a response center; the receiver inverts its own dictionary. Target uncertainty remains inside Δ and may survive full revelation. It also supplies scalable baselines for goal spaces where exact MILP is unnecessary.

Closed form on a noisy cycle

We now obtain an exact frontier rather than a generic approximation. Let $K = 2h + 1$ physical levers lie on a cycle. The sender observes estimate $x \in \mathbb{Z}_K$ while the true target satisfies $d_K(x, t) \leq r$. Receiver type y is a nominal frame shift, but the realized local-to-physical dictionary contains an unobserved bias δ with $d_K(\delta, 0) \leq s$. Local action b selects $b + y + \delta \pmod{K}$, with payoff

$$u(t, b, y, \delta) = 1 - \frac{d_K(b + y + \delta, t)}{h}.$$

The hidden bias models uncertainty about the process producing the receiver's action dictionary; neither agent observes it.

An M -message protocol quantizes sender estimates into at most M intended physical centers. The optimal covering radius of an odd K -cycle is

$$\rho_K(M) = \left\lceil \frac{K - M}{2M} \right\rceil.$$

For an estimate-center distance d , sender and receiver uncertainty dilate the worst physical error to $\min\{h, d + r + s\}$: the triangle inequality gives the upper bound, and moving target and dictionary in opposite directions attains it.

Theorem 11 (Additive private-lever frontier). *For $1 \leq M \leq K$,*

$$\text{EPoC}_{K,r,s}(M) = \frac{\min\{h, \rho_K(M) + r + s\}}{h}.$$

Full sender-type revelation leaves the irreducible floor $\min\{h, r + s\}/h$.

The three terms have distinct meanings: $\rho_K(M)$ is communication quantization, r is target uncertainty seen by the sender, and s is latent receiver-dictionary uncertainty. The receiver removes its observed nominal frame by subtracting y , but no sender message reveals t or δ . For tolerance $\epsilon < 1$, let $R = \lfloor \epsilon h \rfloor - r - s$. If $R < 0$, no message budget attains the tolerance; otherwise

$$C_{K,r,s}(\epsilon) = \left\lceil \frac{K}{2R + 1} \right\rceil.$$

The supplement proves both the cycle covering formula and the exact two-sided uncertainty dilation.

Unknown Model Channels

The world set can also encode uncertainty about how private models were produced. Let a candidate channel Q induce possible worlds Ω_Q and an ambiguity family $\mathcal{Q}$ induce $\Omega_{\mathcal{Q}} = \bigcup_{Q \in \mathcal{Q}} \Omega_Q$. All preceding results apply to the union instance.

Proposition 12 (Ambiguity monotonicity). *If $\mathcal{Q} \subseteq \mathcal{Q}'$, then $\text{EPoC}_{\mathcal{Q}}(M) \leq \text{EPoC}_{\mathcal{Q}'}(M)$ for every M , and $C_{\mathcal{Q}}(\epsilon) \leq C_{\mathcal{Q}'}(\epsilon)$ for every ϵ .*

The statement follows because each protocol is evaluated on a superset of worlds. Its practical implication is less tautological: a protocol optimized for one assumed model channel may be maximally fragile even when robust design substantially reduces the misspecification loss.

The irreducible part has a general cellwise form when the sender's action is trivial. Let $\Omega(x, y)$ be the latent worlds inducing observed type pair (x, y) and define

$$\delta(x, y) = \min_{b \in B} \max_{\omega \in \Omega(x, y)} [v_{\omega}^* - u_{\omega}(b)].$$

Proposition 13 (Full-revelation floor). *In every receiver-only game, if $M \geq |X|$, then*

$$\text{EPoC}(M) = \max_{(x, y): \Omega(x, y) \neq \emptyset} \delta(x, y).$$

The sender can reveal x , after which the receiver chooses one robust action per observed cell (x, y) ; no sender message can distinguish worlds inside the same cell. Hence a flat frontier identifies missing observations rather than an insufficient vocabulary.

For the ambiguous cyclic dictionary, let $K = 2h + 1$ and let D_y contain the unobserved additive biases for receiver type y . Translation invariance gives the corollary

$$\begin{aligned} \text{EPoC}(K) &= \frac{1}{h} \max_y \text{rad}(D_y), \\ \text{rad}(D) &= \min_c \max_{\delta \in D} d_K(c, \delta). \end{aligned}$$

For $D_y = \{0, d_y\}$, this radius is $\lceil d_K(0, d_y)/2 \rceil$. Thus additional sender communication cannot remove receiver-side uncertainty about its own dictionary.

We instantiate this with an ambiguous cyclic action dictionary. The sender knows one of 13 physical targets. Receiver type $y \in \{0, 1, 2, 3\}$ specifies a nominal frame shift, but the true local-to-physical mapping may contain an additional unobserved bias: respectively 0, 2, 4, or 6 positions. The nominal designer assumes zero bias. The robust designer includes both nominal and biased mappings for each frame. This changes neither agents' observed types nor the communication budget; it changes only which game model is considered possible.

Exact Experiments

All experiments use exact SciPy/HiGHS MILPs or theorem-specific algorithms on a single CPU; no policy training is performed. The artifact contains raw tables, 29 unit/cross-check tests, deterministic seeds, and scripts for every figure.

Higher-order separation. For prime $q \in \{2, 3, 5, 7, 11, 13\}$, exact projective-plane hitting sets equal $q + 1$ throughout, whereas the empty pairwise graph predicts one message (Figure 2a). At $q = 13$, 183 private models require 14 messages.

Experiment	Baseline	Exact/rob.	Prediction
Projective plane $q = 13$	1	14	14
Random sets, size 10	1.00	3.00	–
Cycle $r = s = 1, 4$ bits	–	1/3	1/3
5×5 map, $M = 4$	0.50	0.25	0.25
Ambiguous dictionary, 4 bits	1.00	0.50	0.50
Intervals, audit cases	–	20/20	exact

Table 2: Representative deterministic results. Baselines are pairwise coloring, farthest-first, or the best-transferring nominal protocol as appropriate; random-set entries are means over 20 seeds.

Additive cyclic uncertainty. For $K = 13$, two private frames, five (r, s) settings, and zero to four bits, the general protocol MILP matches Theorem 11 on all 25 configurations (Figure 2b). An exhaustive audit over every $r, s \in \{0, \dots, h\}$ and $M \in [K]$ for $K \in \{3, 5, 7\}$ adds 169 exact matches. The $(r, s) = (1, 0)$ and $(0, 1)$ frontiers coincide; with 13 messages, regret is $(r + s)/6$, so communication removes quantization but neither hidden uncertainty source.

Two-dimensional map alignment. We use a 5×5 physical grid with Manhattan payoff and all eight square rotations/reflections as private receiver frames. The exact metric-center EPoC for $M \in \{1, 2, 4, 8, 16, 25\}$ is $\{.50, .375, .25, .125, .125, 0\}$ (Figure 3a). The unrestricted protocol MILP agrees through $M = 8$ on two nontrivial frames; at $M = 4$, repeating it with 1, 2, 4, and all 8 frames always returns .25. Farthest-first reaches its worst-case factor two on the audited 3×3 – 5×5 grids, illustrating both the imported guarantee and the value of exact protocol design on small maps.

Unknown dictionaries. In the asymmetric ambiguity instance (Figure 3b), a nominally optimal full-revelation protocol has zero in-model regret but worst-case regret one over the possible hidden dictionaries. Robust design reduces this to $1/2$, exactly the cellwise minimax floor. More target detail cannot identify an unobserved receiver-side model error.

Generic higher-order gaps and intervals. Across 140 seeded set systems, the true message requirement is 1.55 times the pairwise prediction on average and up to three times larger (Figure 3c); greedy hitting set averages 1.056 times optimum and reaches 1.5 on the worst seed. For intervals, right-endpoint greedy matches the MILP on all 20 audit instances and scales to 10,000 sender types; timing is a hardware-dependent sanity check, while exactness follows from the theorem.

Related Work

Epistemic game theory and teams. Common knowledge formalizes iterated mutual knowledge and has long been central to coordination and agreement [Aumann 1976, Geanakoplos 1994]. Team decision theory studies the joint design of information and decision rules under decentralized observations [Marschak and Radner 1972], while communication equilibria study strategic mediation and pre-play reports in

Bayesian games [Forges 1986]. Our agents have identical payoffs and follow a designed protocol, so incentive compatibility is not the issue. We instead quantify which private descriptions must be distinguished to approximate the world-specific team optimum.

Zero-shot coordination and ad hoc teamwork. Ad hoc teamwork studies cooperation without prior coordination [Stone et al. 2010, Tambe 1997]. Other-Play removes arbitrary symmetry breaking by optimizing cross-play under relabelings [Hu et al. 2020]; later methods train diverse partner populations or open-ended cooperative curricula [Strouse et al. 2021, Li et al. 2023], and recent work targets cross-environment or evolving-game generalization [Jha et al. 2025, Hui et al. 2026]. These methods primarily learn policies. We instead ask an exact design question: given private game descriptions, what worst-case value can a bounded interoperability message certify?

Noisy zero-shot coordination is especially close: agents receive noisy private versions of an underlying Dec-POMDP [Anwar et al. 2024]. Our finite model abstracts away training and allows uncertainty about which worlds or model-generation processes are possible. More importantly, EPoC varies the message budget and benchmarks against a world-aware oracle, exposing a complete repairability frontier.

Coordination and zero-error information theory. Coordination capacity characterizes distributions implementable with communication and common randomness [Cuff et al. 2010, Cuff 2013]. Zero-error side-information coding relates exact recovery to graph coloring [Witsenhausen 1976]; Hide-and-Seek and zero-error coordination study permissible action sets under private observations [McEliece and Posner 1971, Abroshan et al. 2015]. Our one-receiver set-system specialization inherits these classical connections. The distinct emphasis is payoff-sensitive regret over private *game models*, higher-order compatibility across strategically interchangeable types, and structural boundaries linking action cardinality, Helly dimension, and epistemic graph width. In communication-complexity language, $C(\epsilon)$ is a zero-error one-way partition complexity for a payoff-threshold relation. Our compatibility hypergraph makes explicit that the legal cells of this partition can have higher-order, rather than pairwise, obstructions. The metric-dictionary specialization connects protocol compression to classical metric clustering and uncertain-center models [Gonzalez 1985, Xu and Zhang 2025]. We use the farthest-first guarantee of Gonzalez [1985]; the new modeling step identifies when that geometric objective is exactly the oracle-relative coordination loss.

Decentralized control. General decentralized partially observed control is computationally hard [Bernstein et al. 2002]. Rather than approximate a general Dec-POMDP, we isolate a one-shot interoperability layer and characterize exactly when its complexity collapses to 2-SAT, interval stabbing, or bounded-treewidth CSP.

Implications for Multi-Agent Design

Handshake symbols should encode strategic classes. A natural engineering approach is to hash, cluster, or serial-

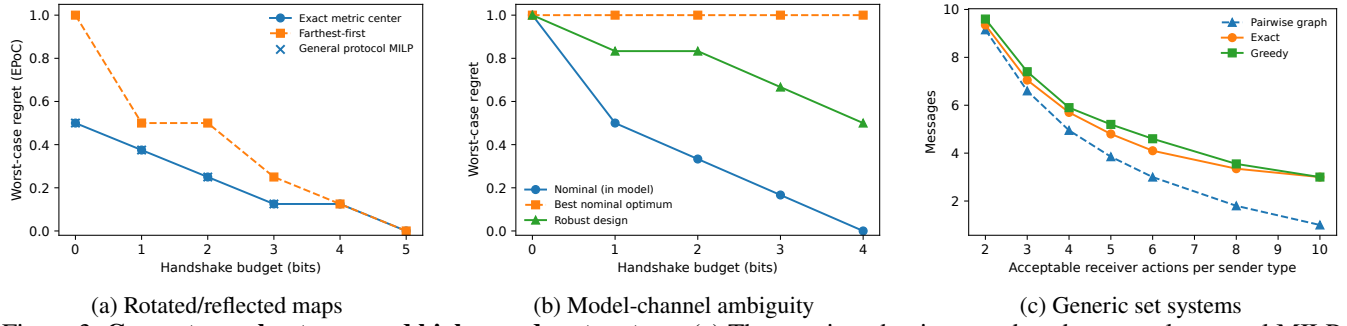


Figure 3: **Geometry, robustness, and higher-order structure.** (a) The metric reduction matches the general protocol MILP; farthest-first can be a factor two worse. (b) Robust design controls an unobserved action dictionary but cannot erase its minimax floor. (c) Pairwise coloring systematically underestimates higher-order message requirements.

ize private model parameters. Theorem 4 gives a different target: a message class is valid only when all of its models admit a common continuation for each receiver type. Consequently, representational similarity is neither necessary nor sufficient. Two numerically different models can share a symbol when their acceptable action relations align, while pairwise-compatible models may still fail collectively.

Minimal incompatibilities are diagnostic certificates. A hyperedge identifies a smallest collection of local models that cannot share a handshake symbol. Such certificates can be returned alongside an infeasible MILP to explain why a protocol vocabulary is too small. Theorem 8 shows when diagnosis is local: in d -dimensional convex response spaces, at most $d + 1$ models witness failure. The projective family shows that no universal pairwise certificate exists without structural assumptions.

A flat frontier calls for sensing, not more language. The additive-cycle, map, and ambiguous-dictionary experiments distinguish loss caused by compressed conventions from loss caused by absent information. Once $\text{EPoC}(M)$ reaches its floor, increasing the message alphabet cannot help; the system must improve sensing, model calibration, message direction, or the information revealed to the other agent. Thus the frontier is also a design diagnostic for deciding whether to allocate resources to communication or to better private models.

Limitations and Conclusion

The present theory is deliberately narrow. It treats two agents, deterministic one-way pre-play messages, finite worst-case world sets, and predominantly synthetic games. Positive-regret randomization, interactive protocols, learned or continuous private models, many-agent coalitions, and Bayesian objectives can change both the operational interpretation and the complexity. Our exact MILP is a validation tool rather than a large-scale solver, and classical hitting-set hardness still applies to an important subclass.

Within this scope, the results give a precise answer to a basic interoperability question. The cost of private game models is a payoff-sensitive staircase, not a binary property. Its communication threshold is governed by higher-order compatibility, which pairwise diagnostics can underestimate

without bound; yet convexity, intervals, and sparse type interactions restore tractability. The closed-form lever family further separates mismatch that a handshake repairs from uncertainty that no message can remove. These distinctions provide a compact theoretical foundation for designing and auditing communication among heterogeneous cooperative agents.

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